SAMPLE

Famous Astronomers: Jan Oort

The Cloud of Comets

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Introduction

The man having been born in the same spring as the century, 1900 in the Frisian town of Franeker, and having died in the century's last decade, 1992 in Leiden, the span between those two dates being nearly the whole of modern astronomy, and the volume being the record of that span. The reader being introduced to the man who found the Galaxy turning, who weighed the disk and found it too light, who hung the comets in a distant sphere, and who brought the radio sky into the Dutch polders, the four being a single career, that career being this volume's whole subject. The book's method being the series' standard: the primary documents having been read, the standard biography having been consulted, the man's own papers having been quoted sparingly and honestly, and the errors having been reported at their full length rather than trimmed for comfort. The chapters walking the house's fixed route, the route being walked at the shelf's unhurried pace, and the pace being justified by the subject's own patience, the patience having been the instrument behind every result the chapters will cross.

The span's own arithmetic being laid out plainly for the reader who wants the spine: the Friser birth in 1900, the Groningen apprenticeship in the star counts, the Leiden post under the professor of celestial mechanics, the rotation proof in 1927, the disk weighing in 1932, the wartime hiding, the hydrogen campaign from 1944, the comet sphere in 1950, the southern observatory built through the 1950s, the radio array completed as the man retired, and the late papers on the largest structures, the dates being given once here and the chapters being free to walk the same road slowly, the slowness being the series' own preference and the preference being honored throughout.

Chapter 1: The Discovery of Jan Oort

The discovery being the making of an astronomer rather than a single find, the making having run from the Frisian childhood through the Groningen lecture hall to the Leiden office where the Galaxy's turning was first read from the starlight, and the chapter being the record of that making. The boy having been born to the physician of Franeker, the household having been bookish and musical, the Friesland sky having been wide and low over the canals, and the boy's arithmetic having been the family's first notice of the gift, the gift being the quick, sure mental calculation that the colleagues would still be remarking on seventy years later. The schools having carried the boy to Groningen, the university having been the home of the greatest living stellar statistician, Jacobus Kapteyn, and the encounter being the discovery's true first event: the seventeen-year-old having wandered into the lectures on the mechanics of starlight and having never wandered out, the professor having built the Groningen school on star counts and proper motions, and the student having absorbed the whole method, the method being the patient arithmetic of many stars, the patience being the trade the boy would practice for seventy years.

The apprenticeship having continued across the Atlantic, the young graduate having spent the year at the Yale observatory with the visiting professorship, the American stay having been brief and useful, and the return having been to Leiden, the Leiden chair being Willem de Sitter's, the assistant's post having been the young man's formal title, and the real education having been the evening seminars where the professor of celestial mechanics argued the Galaxy's structure with anyone who would stay, the structure being the era's open wound, the wound being the question of whether the star streams Kapteyn had counted were two real currents or one system seen from inside. The doctoral thesis having come in 1926, the subject having been the stars of high velocity, the thesis having sorted the fast stars from the slow ones and having noticed what the sorting showed: the fast stars traveling in a preferred direction, the direction being away from the galactic plane's rotation, and the noticing being the seed of everything the man would later prove.

The proof having arrived in 1927, the paper on the observational evidence for the Galaxy's rotation having been the discovery chapter's summit, the argument having been built on the professor Kapteyn's own star counts and the Swede Bertil Lindblad's new theory of a rotating system, and the young Leiden astronomer having done what neither had done: having reduced the motions to two constants, the constants now bearing the man's name, and having shown that the whole swarm of local stars carried one signature, the signature being differential rotation, the inner stars lapping the outer ones like traffic on a ring road. The paper having made the author famous at twenty-seven, the fame having been quiet, the quietness being the man's permanent style, and the style being the discovery's last lesson: the Galaxy having been caught turning by a Frisian who did not raise his voice.

The chapter owing the reader the paper's own mechanics in plain terms: the 1927 argument having rested on the asymmetry in the stars' radial velocities, the velocities having shown the signature of turning when sorted by galactic longitude, the longitude being the angle around the plane, and the pattern having repeated in the proper

motions too, the two independent banks of evidence having been compared and the comparison having closed the case, the case being closed at twenty-seven years of age, the age being the discovery's own arithmetic and the arithmetic being the envy of every later career in the trade.

The American year being given its honest paragraph: the young astronomer having found the Yale post agreeable and the American distances educational, the education having included the continent's scale and the observatories' ambition, the ambition having been carried home and remembered when the European observatory was conceived a quarter-century later, the remembering being the discovery chapter's own transatlantic thread, the thread being documented in the correspondence the archives keep, the keeping being the chapter's documentary warrant.

The chapter closing with the domestic frame: the young astronomer having married the astronomer-delegate's daughter, the marriage having been the long partnership the record shows, the Leiden house having been the career's only permanent address, and the frame being kept because the man's stability having been the method's own footing, the footing having carried the work through two wars and one revolution in technique.

The marriage having been the discovery chapter's own domestic anchor: the young astronomer having wed the daughter of the astronomer the Groningen school sent to the Dutch East Indies, the bride having been the household's organizer and the professor's shield, the shield having been the arrangement that let the man work through the decades the record documents, and the couple having kept the Leiden house for sixty years, the sixty being the discovery's last date and the date being entered here with the respect the record demands.

The discovery closing with the first papers' own reception: the 1927 paper having crossed the trade immediately, the journals' correspondence having shown the era's best minds checking the arithmetic, the checking having confirmed the constants, and the confirmation having made the young Leiden astronomer the trade's point of reference for the Galaxy's mechanics, the reference being kept for sixty years and the sixty being the discovery's own horizon, the horizon being measured from a spring in Franeker and the measurement being complete.

The discovery having also included the instruments the young man built his career on: the meridian instruments of the old statistical school having been the thesis's tools, the tools having been the patient telescopes that measured a star's crossing a thousand times to fix one motion, and the patience of the crossing having been the moral education the era gave its astronomers, the education being transferable to the radio era and the transfer having been the professor's own doing, the doing being the discovery chapter's quietest lesson.